

Chapter 1 Linear Spaces

1.1 Basis

1.1.1 Steinitz's Exchange Lemma

◇ **Lemma 1.1.1.1** (Steinitz) Let U, V be two finite subsets of V . Let U be a set of linear independent vectors. and W spans V . Then,

- $\text{card}(W) \leq \text{card}(U)$;
- There is subset $W' \subseteq W$ with $|W'| = |U|$ and $U \cup W'$ spans V .

◦ **Proof** ...

□

S.Maclane develops this lemma for matroid: